

Note 14.5 Direction Derivatives and Gradient Vectors

Directional Derivatives in the Plane

$$\left(\frac{df}{ds}\right)_{\mathbf{u}, P_0} = \lim_{s \rightarrow 0} \frac{f(x_0 + su_1, y_0 + su_2) - f(x_0, y_0)}{s}$$

where s is the distance increasing through the direction $\mathbf{u}$, $\mathbf{u} = u_1\mathbf{i} + u_2\mathbf{j}$ and the point $P(x_0, y_0)$.

Here, $\mathbf{u}$ includes the direction information and P_0 includes the now position information. the derivative is the rate of change of f in the direction $\mathbf{u}$ and it is the slope of the tangent to the point P_0 in the plane formed by $\mathbf{u}$ and $\mathbf{k}$.

And it can be also denoted by $(D_{\mathbf{u}}f)_{P_0}$.

Calculation and Gradients

Theorem 9--The Directional Derivative Is a Dot Product

$$\left(\frac{df}{ds}\right)_{\mathbf{u}, P_0} = (\nabla f)_{P_0} \cdot \mathbf{u}$$

(only when f is differentiable) where the

$$\nabla f = \frac{\partial f}{\partial x}\mathbf{i} + \frac{\partial f}{\partial y}\mathbf{j} = \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right\rangle.$$

, called the gradient vector, at the point P_0 .

The dot product can be obtained by rewrite the directional derivatives through the chain rule or through the definition and the differentiation.

The gradient vector includes the information of P_0 and it is static at P_0 and the $\mathbf{u}$ includes the information of direction.

Properties of the Directional Derivative

$$D_{\mathbf{u}}f = \nabla f \cdot \mathbf{u} = |\nabla f| \cos \theta$$

1. The direction of ∇f is the direction that the function f increases most rapidly. ($\theta = 0$)
2. Decreases most rapidly in the direction $-\nabla f$. ($\theta = \pi$)
3. Zero change when $\mathbf{u}$ orthogonal to the gradient vector and it is the direction of the tangent line to the level curve. ($\theta = \frac{\pi}{2}$)

Gradients and Tangents to Level Curves

At every point in the domain of a differentiable function, the gradient is normal to the level curve through (x_0, y_0) .

It can be obtained by take derivatives of the both sides of

$$f(g(t), h(t)) = c$$

Algebra Rules for Gradients

Sum rule, difference rule, constant multiple rule, product rule, quotient rule(similar to derivatives).

Functions of Three Variables

Similar to two-variable functions and just add a term.

Exercises

P808 13 29 33